Proxmox, pfSense, Ubuntu DNS, Tailscale, and VLAN Network Deployment

Detailed Implementation Documentation

Secure home and IoT network segmentation using virtualization, pfSense firewalling, internal DNS, Tailscale remote access, managed switching, and VLAN-aware wireless isolation.

Item	Details
Project focus	Secure segmented home/IoT lab behind an existing upstream router
Core platforms	Proxmox VE, pfSense CE, Ubuntu Server / Technitium DNS, TP-Link ES205G switch, TP-Link EAP access point, Tailscale
Security controls	VLAN segmentation, interface-based firewall policy, internal DNS, DHCP scopes, management VLAN, Tailscale subnet routing, and configuration backups
Content scope	Credentials, device secrets, auth keys, public IP details, and private account identifiers are intentionally omitted.

Prepared by: Asadul Islam Arif

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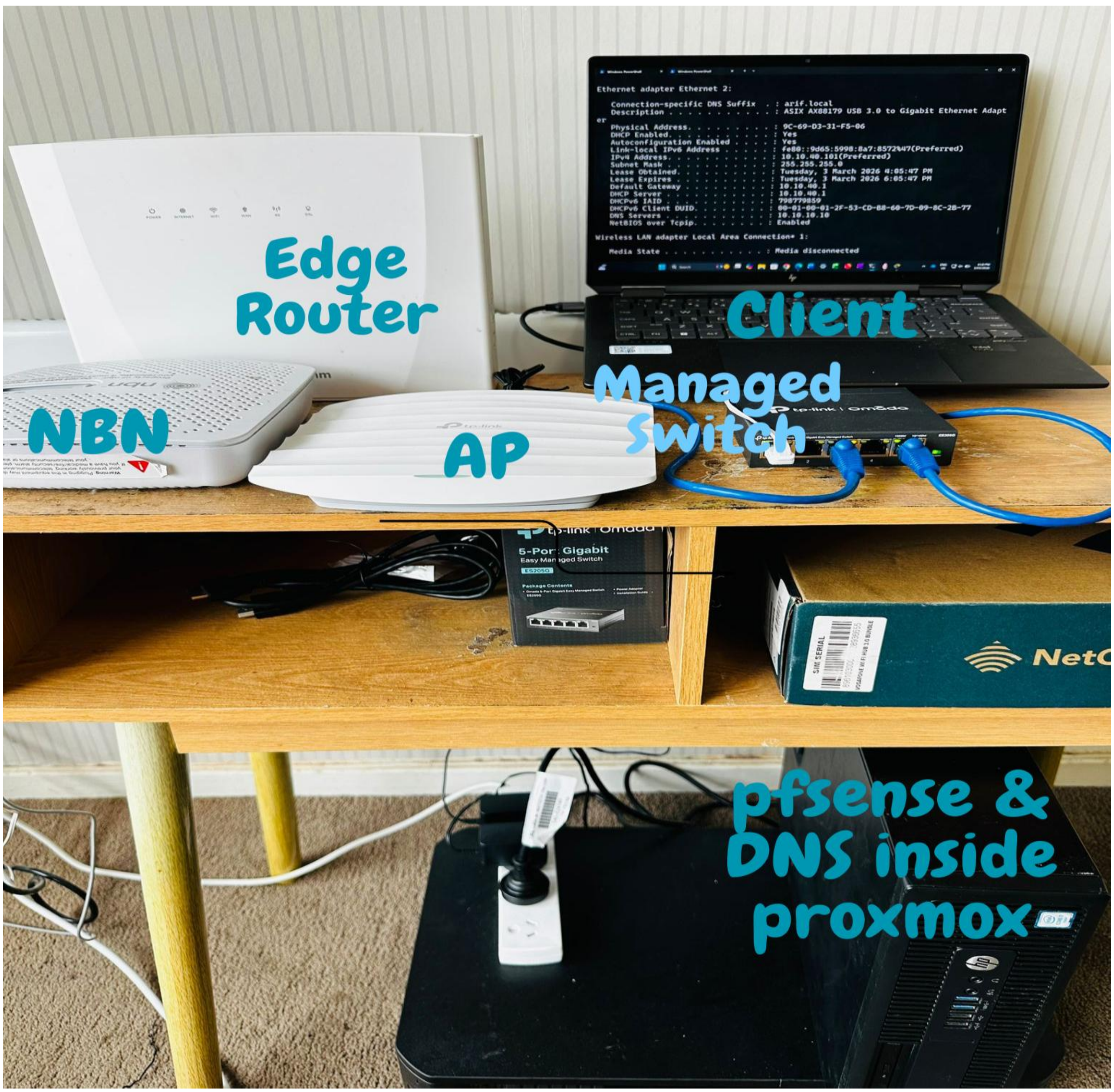
Contents

1. Executive Summary

This document describes the implementation of a segmented home and IoT network built on a Proxmox VE host, with pfSense operating as the main virtual firewall and router. The environment sits behind an existing NBN/upstream home router instead of replacing it. This keeps the normal household internet path stable while allowing the lab network to operate as a separate, firewall-controlled security domain.

Remote administration uses Tailscale rather than a traditional inbound VPN. Both the administrator laptop and pfSense join the same tailnet, and pfSense advertises selected internal routes into it. This provides secure remote access without exposing pfSense or any internal service through public inbound port forwarding on the upstream router, and the connection continues to work when the public IP changes.

The design separates management, NAS, IoT, normal users, guests, and cameras into distinct VLANs. Ubuntu Server provides internal DNS through Technitium DNS, while pfSense provides DHCP, VLAN gateways, routing, NAT, and firewall enforcement. A TP-Link managed switch carries the VLANs over trunk and access ports, and a TP-Link access point maps wireless SSIDs to the correct VLAN IDs.



Proxmox, pfSense, Ubuntu DNS, Tailscale — VLAN Network Implementation

Figure 1: Physical lab setup showing the NBN/upstream router, Proxmox host, TP-Link managed switch, TP-Link access point, and client device used for management and testing.

2. Design Rationale and Security Model

The design enforces a simple security principle: management services are not broadly reachable from every network. Administrative devices are grouped in a dedicated management VLAN (VLAN 99) and are reachable locally only from that segment, or remotely through Tailscale. Normal user, guest, IoT, and camera traffic is separated into independent VLANs, so a compromised or untrusted device has no direct path to administrative services.

2.1 Why pfSense was used

- pfSense provides one central point for VLAN gateways, DHCP scopes, firewall policy, NAT, DNS forwarding options, and routing decisions.
- Each VLAN is treated as a separate security zone with its own interface rule set.
- Rules are expressed through named aliases such as DNS_SERVER, MANAGEMENT_HOSTS, RFC1918_PRIVATE, and MGMT_PORTS, which keeps the rule sets readable and consistent across interfaces.

2.2 Why Tailscale was chosen

Tailscale was chosen as the remote-access layer over traditional VPN options such as WireGuard or OpenVPN, because it does not require any inbound port to be exposed on the WAN side. A self-hosted VPN typically depends on a public IP or dynamic DNS, a port forward on the upstream router, and a matching WAN firewall rule. Tailscale instead establishes outbound connections from both endpoints into the tailnet and falls back to its relay infrastructure when direct peer-to-peer is not possible. The result is a setup with no upstream router changes, no dependency on a static public IP, and a smaller administrative surface at the network edge.

- No UDP port forwarding is configured on the upstream router.
- Only selected routes are advertised to the tailnet: 10.10.99.0/24 and 192.168.1.0/24.
- Remote access is scoped to management, DNS, switch, AP, and Proxmox administration. User, guest, IoT, NAS, and camera VLANs are not reachable through Tailscale.
- Additional VLANs can be advertised through the same pfSense Tailscale configuration if operational requirements change.

3. Network Topology

The physical and virtual topology is structured as an upstream network feeding the Proxmox host, with pfSense bridging the upstream side to the protected VLAN trunk side. The Proxmox host has a bridge for upstream connectivity and a separate internal VLAN-aware bridge for pfSense LAN, Ubuntu DNS, and the physical switch.

Component	Role	Key address / note
Upstream home router	Provides internet to the house and the pfSense WAN side	192.168.1.1/24 network
Proxmox host	Virtualization platform hosting pfSense and Ubuntu DNS	192.168.1.50
pfSense VM	Virtual firewall, router, DHCP server, VLAN gateway, NAT, Tailscale subnet router	WAN 192.168.1.3; management gateway 10.10.99.1
Ubuntu DNS / Technitium	Internal DNS resolver and DNS zone host	10.10.99.10
TP-Link ES205G switch	Managed switch carrying VLAN trunks and access ports	10.10.99.20
TP-Link EAP access point	Wireless VLAN mapping for SSIDs	10.10.99.21
Tailscale	Remote administration overlay	Advertises 10.10.99.0/24 and 192.168.1.0/24

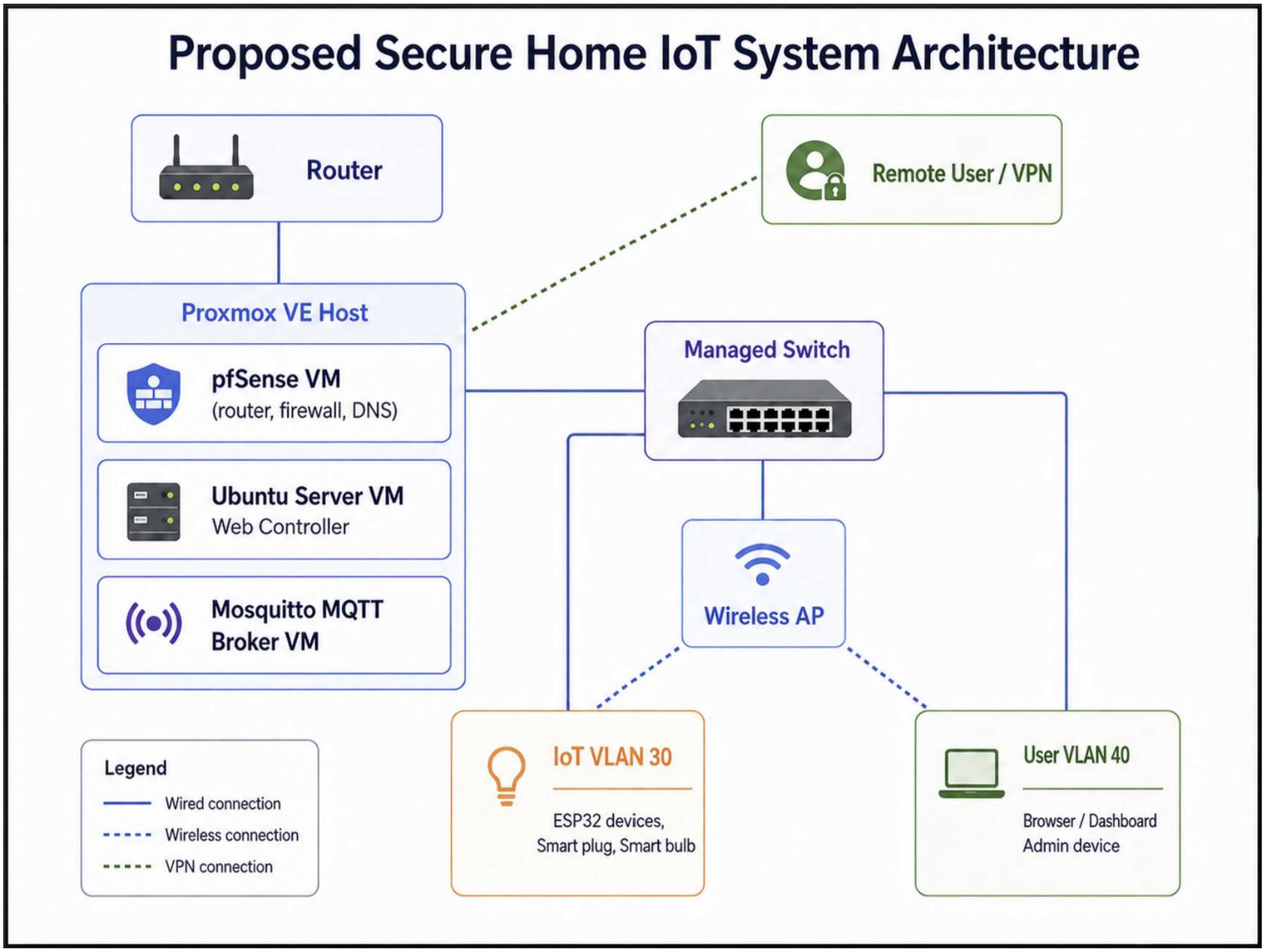


Figure 2: High-level topology showing the upstream router, Proxmox host, pfSense VM, Ubuntu DNS server, managed switch, AP, VLANs, and Tailscale remote management path.

4. IP Addressing, VLAN, and DHCP Plan

The addressing plan separates the native LAN (VLAN 1) from the management VLAN (VLAN 99). VLAN 99 hosts administration and infrastructure devices (pfSense, DNS, switch, AP, and the admin laptop), and VLAN 1 is kept as the default LAN segment.

Segment / purpose	VLAN	Subnet / gateway	DHCP / static plan	Notes
Native LAN	1	10.10.10.0/24, gateway 10.10.10.1	Client pool 10.10.10.100-199 if enabled	Default internal LAN; not used as the primary management VLAN
NAS	20	10.10.20.0/24, gateway 10.10.20.1	Pool 10.10.20.100-199	NAS and storage segment
IoT	30	10.10.30.0/24, gateway 10.10.30.1	Pool 10.10.30.100-199	IoT devices; blocked from private internal networks except allowed services
Normal users	40	10.10.40.0/24, gateway 10.10.40.1	Pool 10.10.40.100-199	General user network
Guest users	50	10.10.50.0/24, gateway 10.10.50.1	Pool 10.10.50.100-199	Internet-only guest access
Cameras	60	10.10.60.0/24, gateway 10.10.60.1	Pool 10.10.60.100-199	Camera network; restricted and local-first

Management	99	10.10.99.0/24, gateway 10.10.99.1	Admin pool 10.10.99.100-199; infrastructure static/reserved	Trusted admin network for pfSense, DNS, switch, AP, and admin laptop
Upstream / Proxmox	-	192.168.1.0/24	Controlled by upstream router	Used for pfSense WAN and Proxmox host management
Tailscale overlay	-	100.x.x.x tailnet addresses	Managed by Tailscale	Remote management overlay; not a replacement for local VLAN addressing

Important static addresses used in the management design:

Device	Address	Reason
pfSense VLAN99 gateway	10.10.99.1	Default gateway and administrative pfSense address for the management VLAN
Ubuntu DNS / Technitium	10.10.99.10	Stable internal DNS address used by DHCP and firewall aliases
TP-Link switch	10.10.99.20	Switch management address
TP-Link AP	10.10.99.21	AP management address
Proxmox host	192.168.1.50	Host management address on upstream/private router network
pfSense WAN	192.168.1.3	WAN-side address behind the upstream router

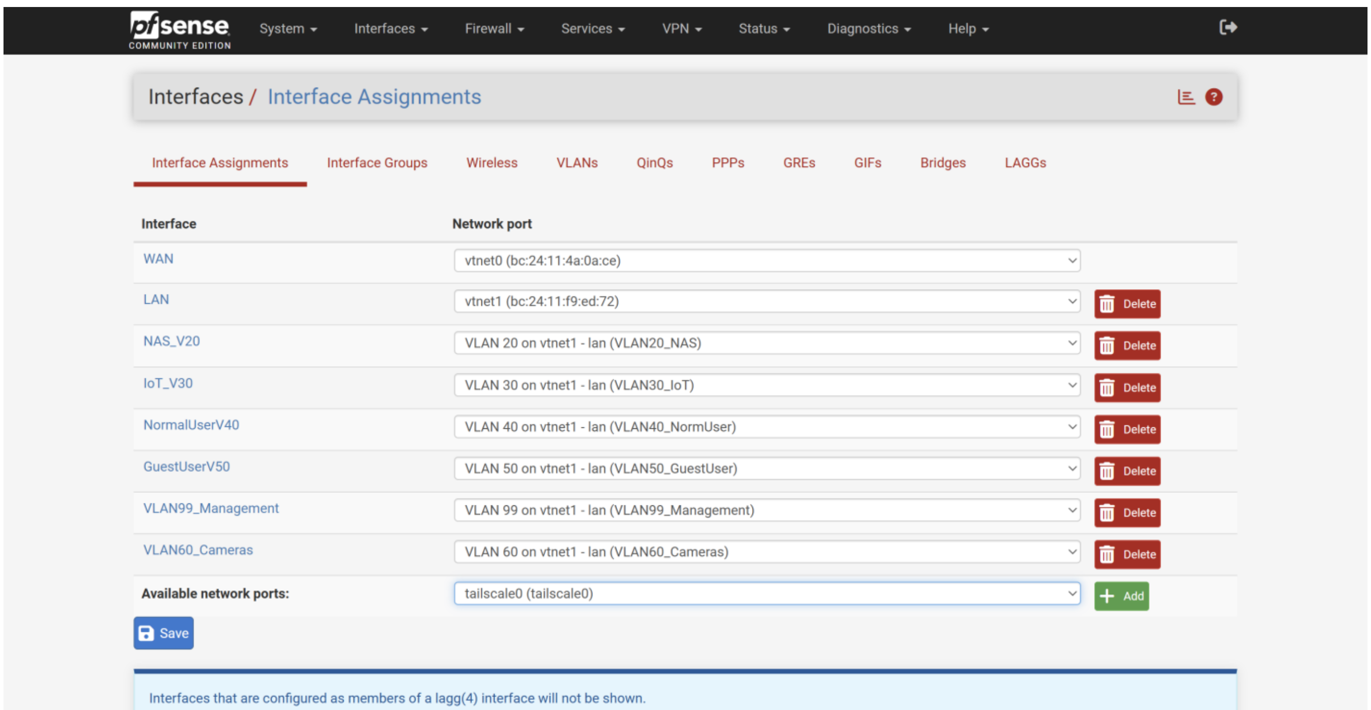


Figure 3: pfSense interface assignment page showing WAN, LAN, Tailscale, NAS_V20, IOT_V30, NORMALUSERV40, GUESTUSERV50, VLAN60_CAMERAS, and VLAN99_MANAGEMENT.

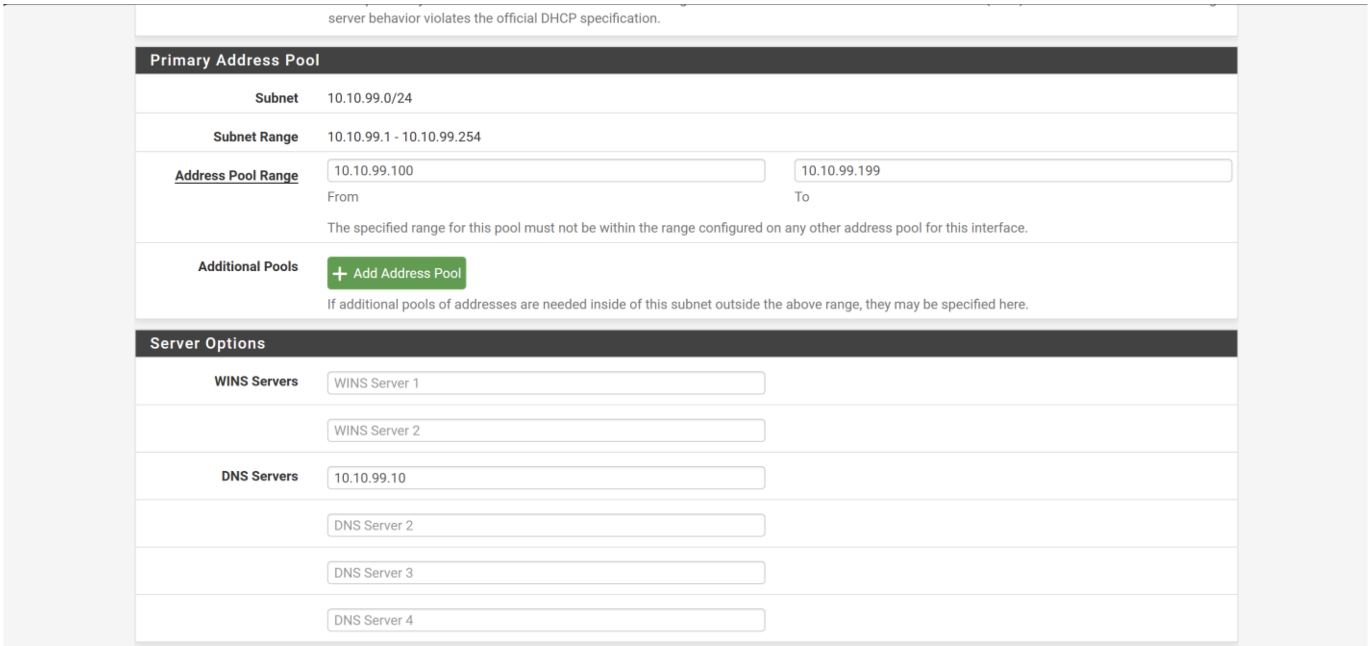


Figure 4: pfSense DHCP Server page for VLAN99_MANAGEMENT showing the management subnet, DHCP pool, and DNS server value 10.10.99.10.

5. Proxmox Host and Virtual Networking

Proxmox VE runs the firewall and DNS virtual machines. The key design decision is to keep upstream access and the protected internal VLAN trunk separate. This prevents the internal VLAN design from depending on the upstream router for segmentation.

5.1 Virtual bridge model

- vmbr0 is connected to the upstream/home router side and is used for the Proxmox management IP and the pfSense WAN interface.
- vmbr1 is connected to the internal/protected side and is used by pfSense LAN, Ubuntu DNS, and the physical switch trunk path.
- VLAN-aware mode is enabled on vmbr1 so that VLAN tags can pass between pfSense, the switch, and the access point.
- pfSense VM uses one virtual NIC on vmbr0 for WAN and one virtual NIC on vmbr1 for the LAN/VLAN trunk side.

5.2 VM layout

VM ID / component	Purpose	Network placement
100 / pfSense	Firewall, router, DHCP, VLAN gateways, Tailscale subnet router	WAN NIC on vmbr0; LAN/trunk NIC on vmbr1
101 / Ubuntu Server	Technitium DNS and utility services	Management VLAN address 10.10.99.10 behind pfSense
Proxmox host	Virtualization management platform	192.168.1.50 on upstream private network

5.3 Proxmox virtual NIC firewall on the pfSense trunk

The Proxmox virtual NIC firewall is left disabled on the pfSense trunk NIC. Enabling Proxmox-level filtering on this interface interferes with VLAN DHCP traffic reaching pfSense, so all VLAN-level filtering is handled by pfSense interface rules. This gives a single, predictable enforcement point for VLAN security and avoids two layers of filtering that could otherwise mask each other’s effects.

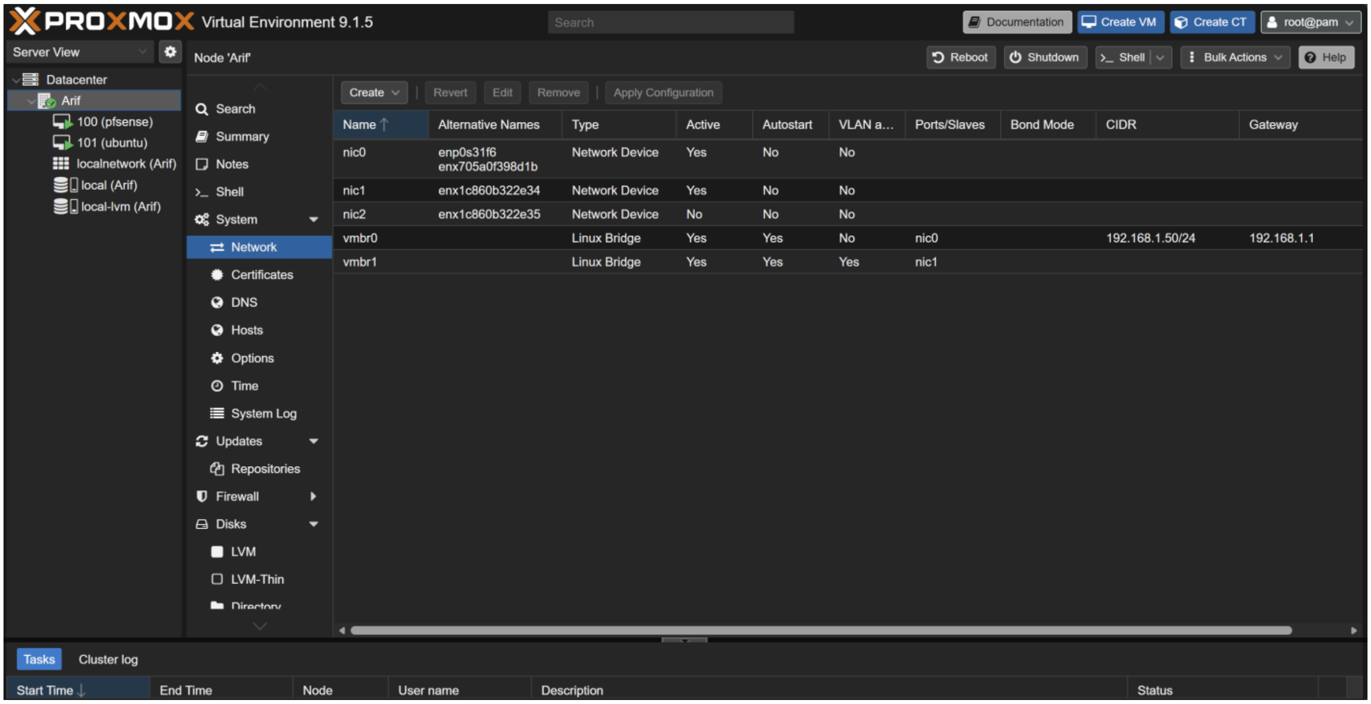


Figure 5: Proxmox network page showing vmbr0 for upstream connectivity, vmbr1 as the internal VLAN-aware bridge, and the physical NICs attached to the correct bridges.

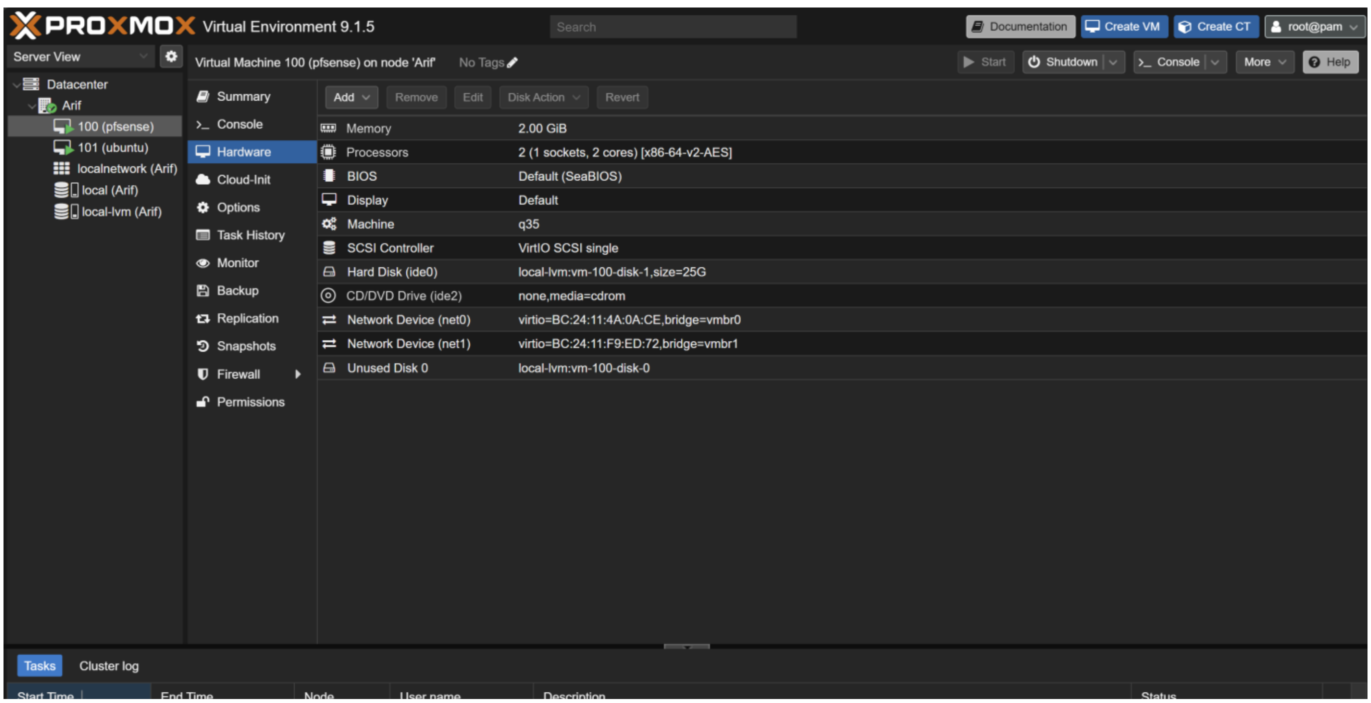


Figure 6: pfSense VM hardware page showing the WAN NIC attached to vmbr0 and the LAN/trunk NIC attached to vmbr1.

6. pfSense Interface, VLAN, and Routing Configuration

pfSense is the core network policy engine. It owns the VLAN gateways, DHCP scopes, NAT to the upstream router, firewall rules, and Tailscale subnet routing. The WAN interface receives upstream connectivity from the 192.168.1.0/24 private network, while the LAN parent interface carries the internal VLANs.

6.1 Interface configuration summary

pfSense interface	Parent / type	Address / subnet	Purpose
WAN	vtnet0 / vmbr0	192.168.1.3/24 via upstream router	Internet uplink behind the NBN/upstream router
LAN	vtnet1 / vmbr1	10.10.10.1/24	Native/default internal LAN
NAS_V20	VLAN 20 on LAN parent	10.10.20.1/24	NAS and storage network
IOT_V30	VLAN 30 on LAN parent	10.10.30.1/24	IoT network
NORMALUSERV40	VLAN 40 on LAN parent	10.10.40.1/24	General users
GUESTUSERV50	VLAN 50 on LAN parent	10.10.50.1/24	Guest users
VLAN60_CAMERAS	VLAN 60 on LAN parent	10.10.60.1/24	Camera network
VLAN99_MANAGEMENT	VLAN 99 on LAN parent	10.10.99.1/24	Infrastructure and admin management
Tailscale	Tailscale package/interface	100.x.x.x tailnet address	Remote management overlay

6.2 WAN exposure policy

The WAN policy is intentionally minimal. No SSH, web GUI, or Technitium port forward is configured on the WAN side, and the upstream router carries no inbound port forwards toward pfSense. Remote administration is handled by Tailscale through outbound connections from both endpoints into the tailnet, so no inbound exposure on the public side is required.

WAN item	Status	Reason
pfSense GUI from WAN	Not exposed	Administrative access occurs from VLAN99 or Tailscale
Ubuntu SSH port forward	Not required	Remote admin can use Tailscale route to management network if needed
Technitium GUI port forward	Not required	Accessed from VLAN99 or Tailscale
Tailscale	Outbound client connection	No inbound WAN rule or router port forward is required

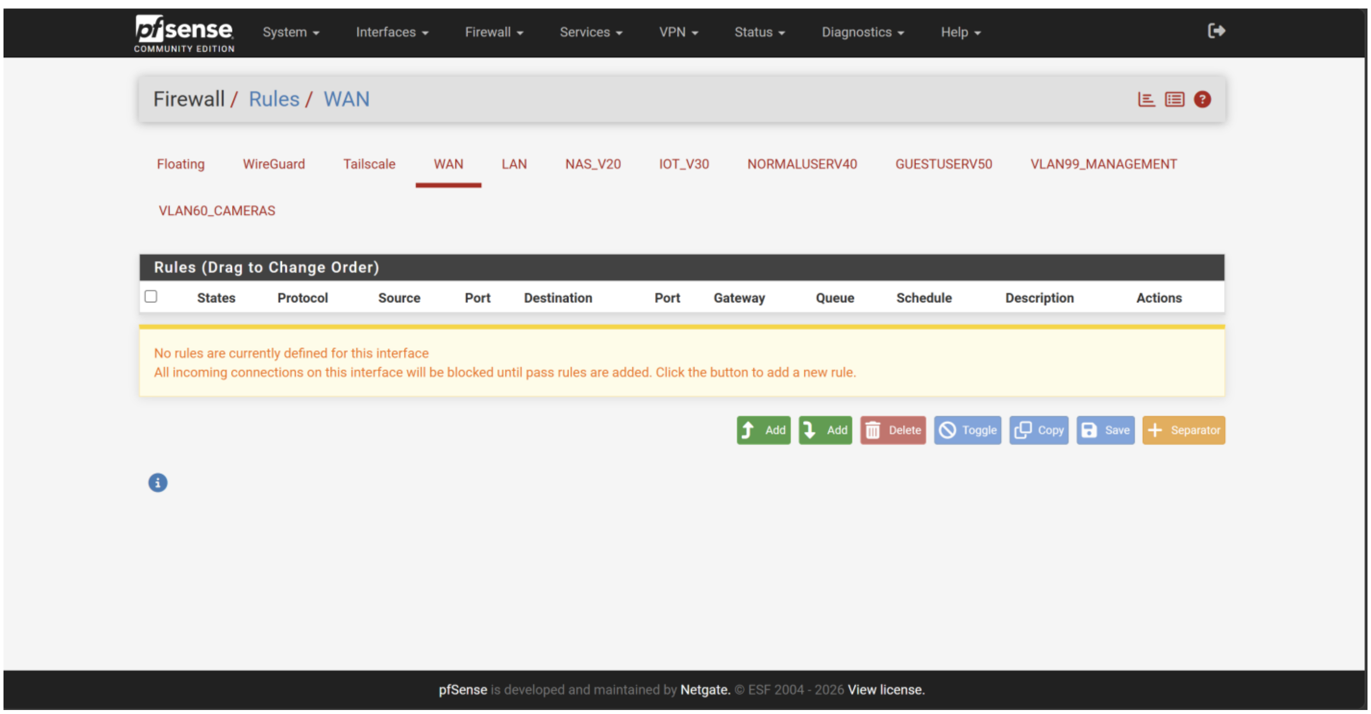


Figure 7: pfSense WAN firewall tab showing no unnecessary public management exposure.

7. Tailscale Remote Access Design

Tailscale is the remote access layer for the environment. It runs on pfSense and on the administrator laptop, with both endpoints joined to the same tailnet. pfSense operates as a subnet router and advertises selected internal networks, so the laptop reaches management services across the tailnet without any service being exposed through the upstream router.

7.1 pfSense Tailscale settings

Setting	Value / approach	Purpose
Enable Tailscale	Enabled	Starts the Tailscale service on pfSense
Authentication	Tailscale account / pre-authentication key	Registers pfSense as a machine in the tailnet
Listen port	Default 41641	Used by Tailscale peer traffic where possible
Accept DNS	Allowed/kept default if needed	Allows tailnet DNS behaviour where appropriate; internal DNS is still 10.10.99.10
Advertise exit node	Disabled	pfSense is not used as a general internet exit node
Accept subnet routes	Disabled	pfSense does not need to accept routes from other Tailscale devices
Advertised routes	10.10.99.0/24 and 192.168.1.0/24	Provides remote access to management and Proxmox/upstream management networks only

7.2 Route scoping rationale

Two networks are advertised to the tailnet. The 10.10.99.0/24 management VLAN provides access to pfSense, DNS, switch, and AP administration interfaces, and the 192.168.1.0/24 upstream/private network provides access to Proxmox host management. The user, guest, IoT, NAS, and camera VLANs are deliberately not advertised, since remote administration of those segments is not part of the current scope. Additional VLAN routes can be advertised through the same pfSense Tailscale configuration and approved in the Tailscale admin console if the scope changes.

7.3 Tailscale access targets

Remote management target	Address	Access method
pfSense management GUI	https://10.10.99.1	Tailscale route to VLAN99
Technitium DNS GUI	http://10.10.99.10:5380	Tailscale route to VLAN99
TP-Link switch GUI	http://10.10.99.20	Tailscale route to VLAN99
TP-Link AP GUI	http://10.10.99.21	Tailscale route to VLAN99; AP UI may load slower over VPN
Proxmox web GUI	https://192.168.1.50:8006	Tailscale route to upstream/private management network

7.4 Tailscale firewall rules

On the Tailscale firewall rules tab in pfSense, explicit rules are used rather than relying on broad implicit behaviour. Tailscale clients are permitted to reach the management hosts on administrative ports and the internal DNS server on port 53; all other traffic is denied. This keeps the remote-access surface aligned with the route-scoping decision in section 7.2.

Order	Action	Protocol	Source	Destination	Port	Description
1	Pass	IPv4 TCP	Tailscale / any trusted tailnet source	MANAGEMENT_HOSTS	MGMT_PORTS	Allow Tailscale admin access to infrastructure devices
2	Pass	IPv4 TCP/UDP	Tailscale / any trusted tailnet source	DNS_SERVER	53	Allow DNS resolution through internal DNS
3	Block/implicit deny	Any	Any	Any	Any	Do not allow unnecessary access to other VLANs

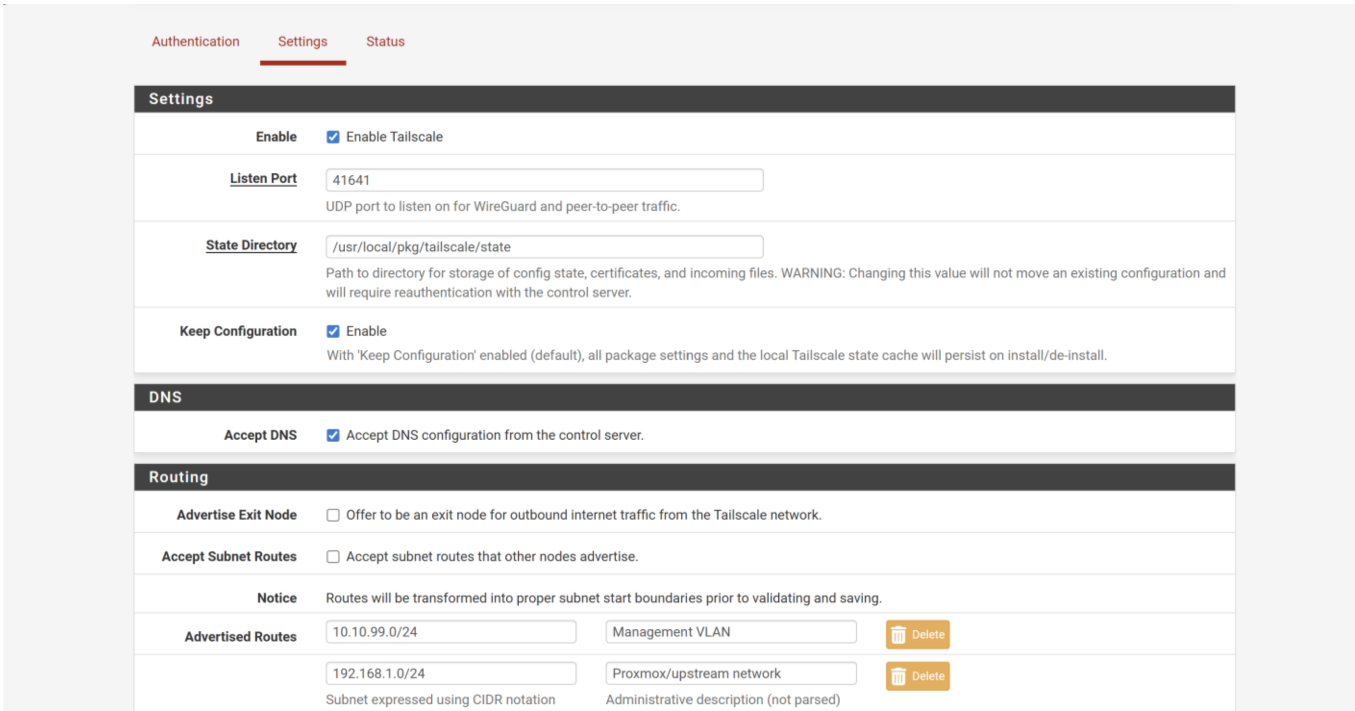


Figure 8: pfSense Tailscale settings page showing Tailscale enabled, exit node disabled, and advertised subnet routes 10.10.99.0/24 and 192.168.1.0/24.

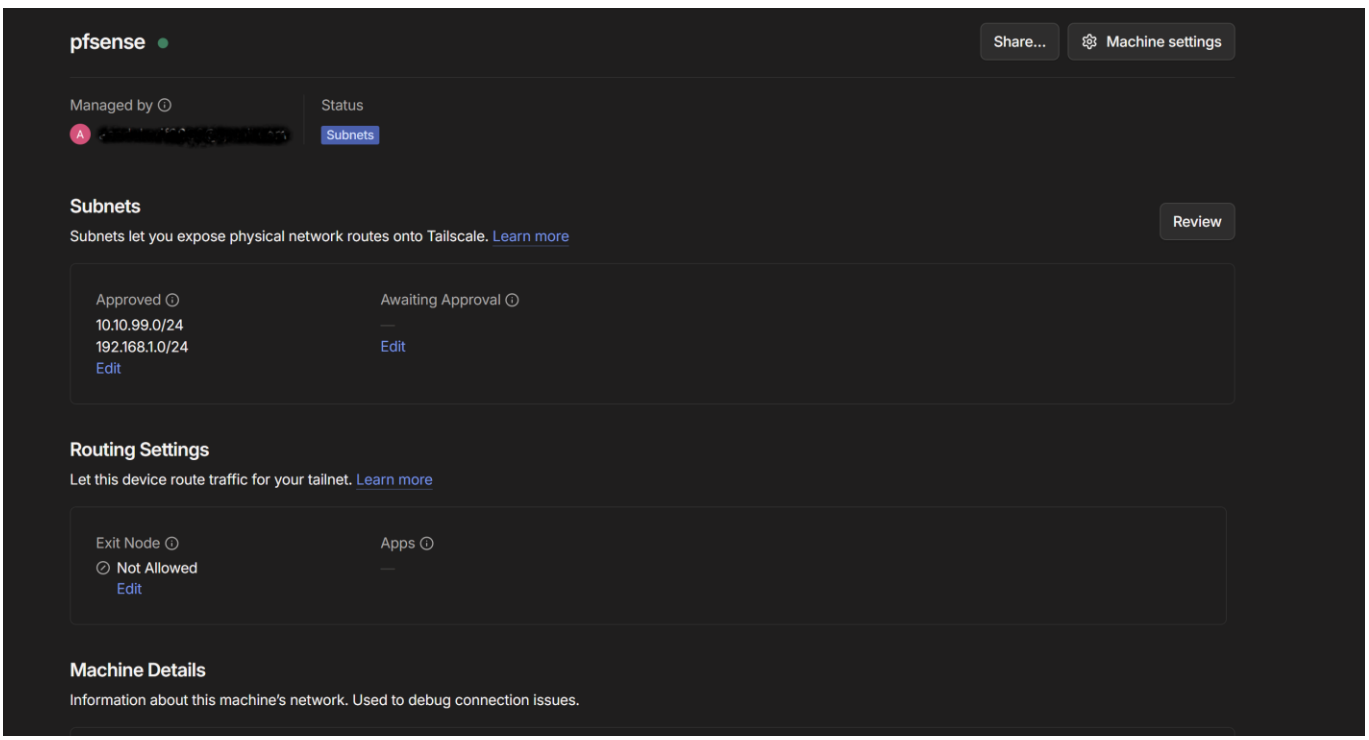


Figure 9: Tailscale admin console showing pfSense online and subnet routes approved for 10.10.99.0/24 and 192.168.1.0/24.

8. Ubuntu DNS / Technitium Configuration

Ubuntu Server provides internal DNS using Technitium DNS. Moving the DNS server into VLAN99 gives it a stable management address and keeps infrastructure services grouped with the other administrative devices. pfSense DHCP scopes point clients toward 10.10.99.10 so local hostnames resolve consistently across the internal environment.

8.1 DNS server addressing

- Ubuntu Server / Ubuntu DNS / Technitium address: 10.10.99.10/24
- Default gateway: 10.10.99.1
- DNS service listens for internal clients as permitted by pfSense firewall policy.
- The DNS address is referenced in the DNS_SERVER pfSense alias and in DHCP server options.

8.2 Internal DNS records

Record	Type	Address	Purpose
dns1.arif.local	A	10.10.99.10	Internal DNS server/Ubuntu Server
pfsense.arif.local	A	10.10.99.1	pfSense management address
switch.arif.local	A	10.10.99.20	TP-Link switch management
ap.arif.local	A	10.10.99.21	TP-Link AP management
proxmox.arif.local	A	192.168.1.50	Proxmox host management

The record names do not need to match the manufacturer device names exactly. However, clear names such as switch, ap, dns1, pfsense, and proxmox are better than random names because they make troubleshooting and documentation easier.

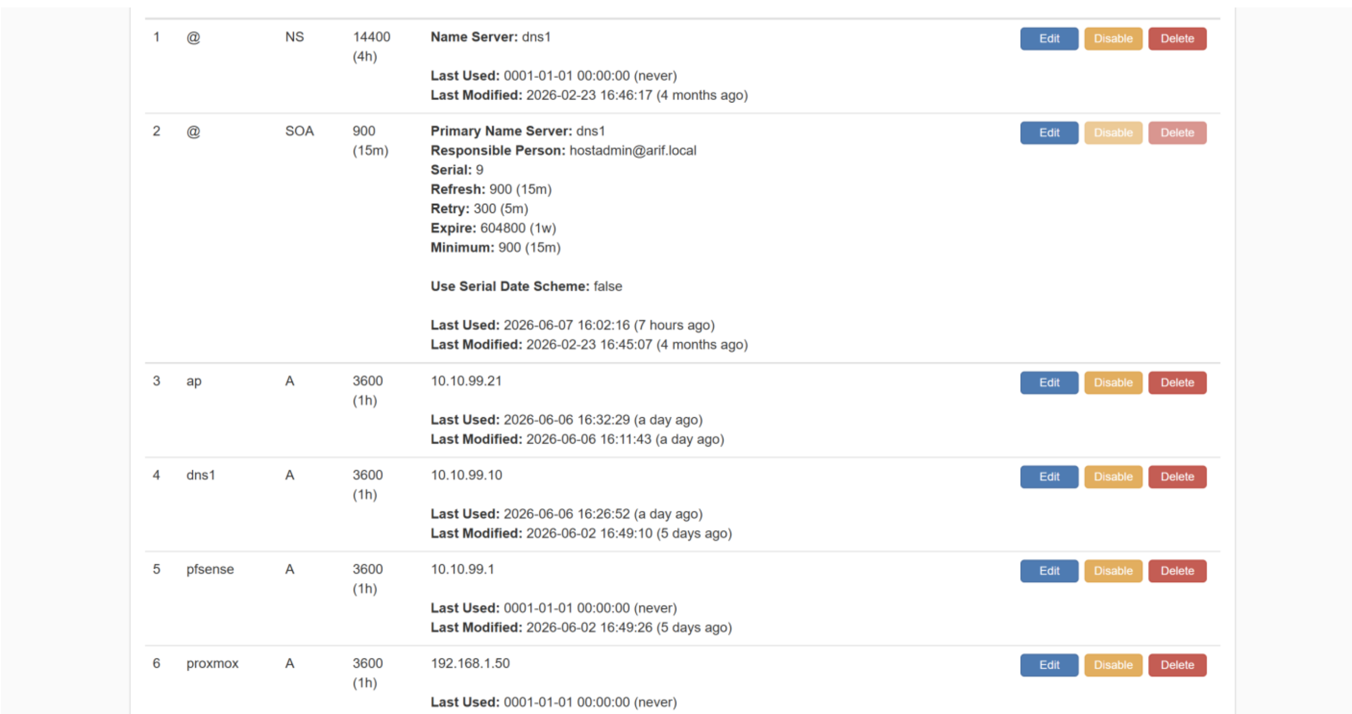


Figure 10: Technitium DNS arif.local zone showing A records, plus the NS and SOA records for the zone.

Figure 11: Windows command prompt or PowerShell showing nslookup queries resolving, switch.arif.local, ap.arif.local

9. Managed Switch Configuration

The TP-Link ES205G switch extends the VLAN design from the Proxmox/pfSense trunk to wired devices and the access point. The switch uses 802.1Q VLANs. Tagged ports are used where multiple VLANs must travel over the same cable, while untagged ports are used for ordinary devices that do not tag traffic.

9.1 Switch IP and management VLAN

- Switch management IP: 10.10.99.20
- Subnet mask: 255.255.255.0
- Default gateway: 10.10.99.1
- DNS server: 10.10.99.10
- Switch management VLAN ID: 99

9.2 Port design

Port	Role	Tagged VLANs	Untagged VLAN / PVID	Reason
Port 1	Trunk/uplink to Proxmox/pfSense internal side	20, 30, 40, 50, 60, 99	VLAN 1 / PVID 1	Carries all routed VLANs to pfSense
Port 2	Physical admin access port	None	VLAN 99 / PVID 99	Allows a laptop to join management VLAN directly by cable
Port 3	Trunk to TP-Link AP	20, 30, 40, 50, 60, 99	VLAN 1 / PVID 1	Carries wireless SSID VLANs and AP management VLAN
Port 4	IoT wired access	None	VLAN 30 / PVID 30	For untagged wired IoT device testing
Port 5	Normal user wired access	None	VLAN 40 / PVID 40	For untagged general user device testing

9.3 Tagged, untagged, member, and PVID logic

- Tagged means the switch sends or receives frames with an 802.1Q VLAN tag. This is used on trunk links such as the pfSense uplink and AP uplink.
- Untagged means frames leave the port without a VLAN tag. This is used for normal end devices that do not understand VLANs.
- Member means the port participates in that VLAN. A port must be a member of a VLAN before it can carry traffic for that VLAN.
- PVID defines which VLAN untagged incoming traffic is placed into. For example, Port 2 with PVID 99 places an ordinary laptop into VLAN99 without needing laptop-side VLAN tagging.

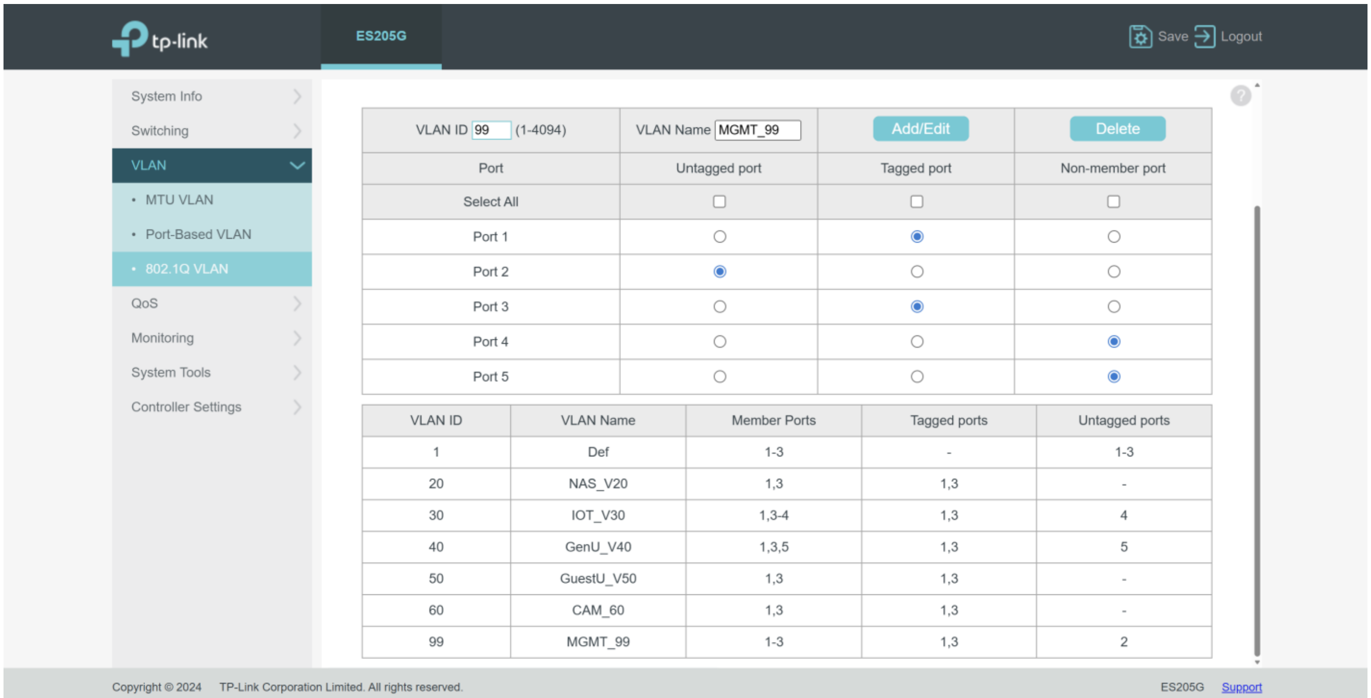


Figure 12: TP-Link ES205G 802.1Q VLAN table showing VLANs 1, 20, 30, 40, 50, 60, and 99, including tagged trunk ports and untagged access ports.

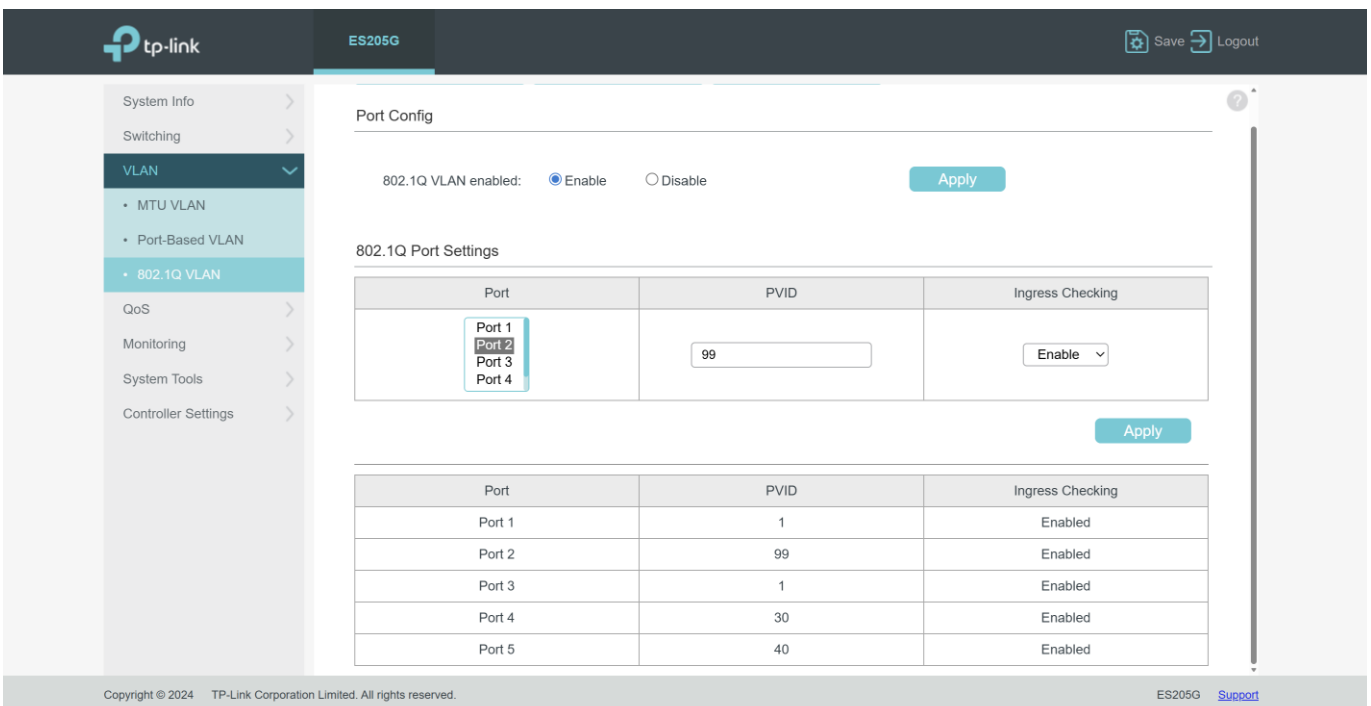


Figure 13: TP-Link ES205G port configuration showing Port 2 PVID 99 for admin access, Port 4 PVID 30 for IoT, and Port 5 PVID 40 for normal user access.

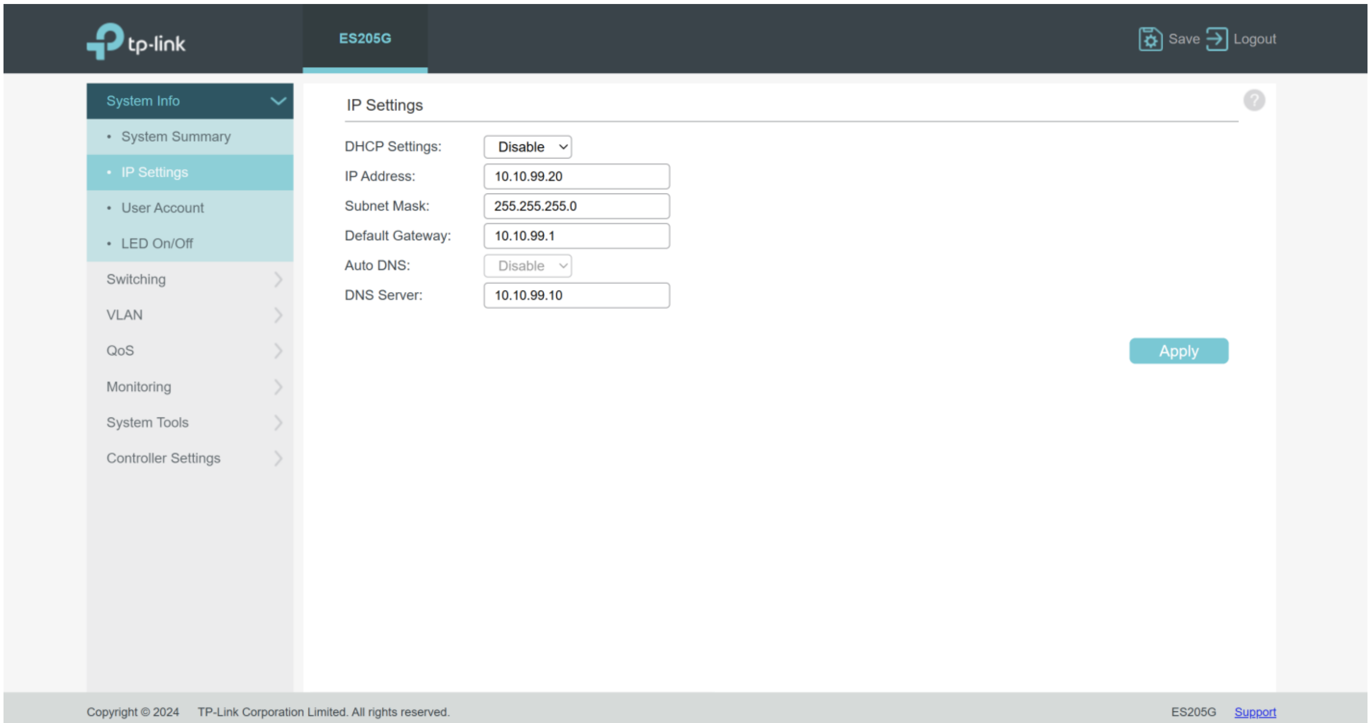


Figure 14: TP-Link switch system information or IP settings page showing management address 10.10.99.20, gateway 10.10.99.1, and DNS 10.10.99.10.

10. Access Point and Wireless VLAN Mapping

The TP-Link access point is connected to the switch on a trunk-style port. It receives management access on VLAN99 and maps individual wireless SSIDs to the correct VLAN ID. This keeps wireless clients separated by purpose without needing separate physical access points.

10.1 AP management configuration

Setting	Value
AP management IP	10.10.99.21
Subnet mask	255.255.255.0
Default gateway	10.10.99.1
Primary DNS	10.10.99.10
Management access policy	Allowed from VLAN99 and Tailscale remote admin path only

10.2 Wireless SSID to VLAN mapping

SSID / purpose	VLAN setting	Expected subnet	Security intent
Default/native SSID	Disabled or unused	N/A or native only if intentionally enabled	Avoids accidental clients on the wrong network
NAS	20	10.10.20.0/24	Storage-related wireless access if required
IoT	30	10.10.30.0/24	Isolated IoT wireless devices
GenU	40	10.10.40.0/24	Normal user wireless devices
GuestU	50	10.10.50.0/24	Guest-only internet access
Cameras	60	10.10.60.0/24	Camera devices with restricted internal access
MgmtU	99	10.10.99.0/24	Admin wireless access only

The AP management IP and the SSID VLAN IDs are independent settings. The AP itself is reached at 10.10.99.21 on the management VLAN, while wireless clients on each SSID are placed into VLAN 20, 30, 40, 50, 60, or 99 according to the SSID's VLAN ID configuration.

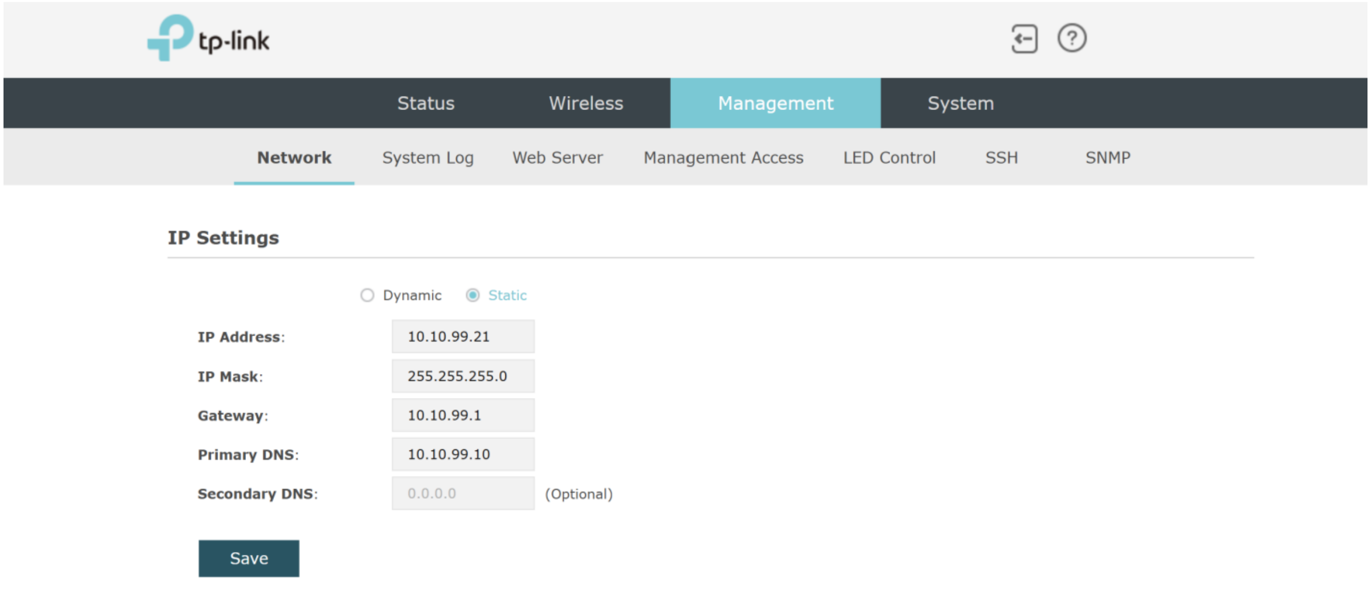
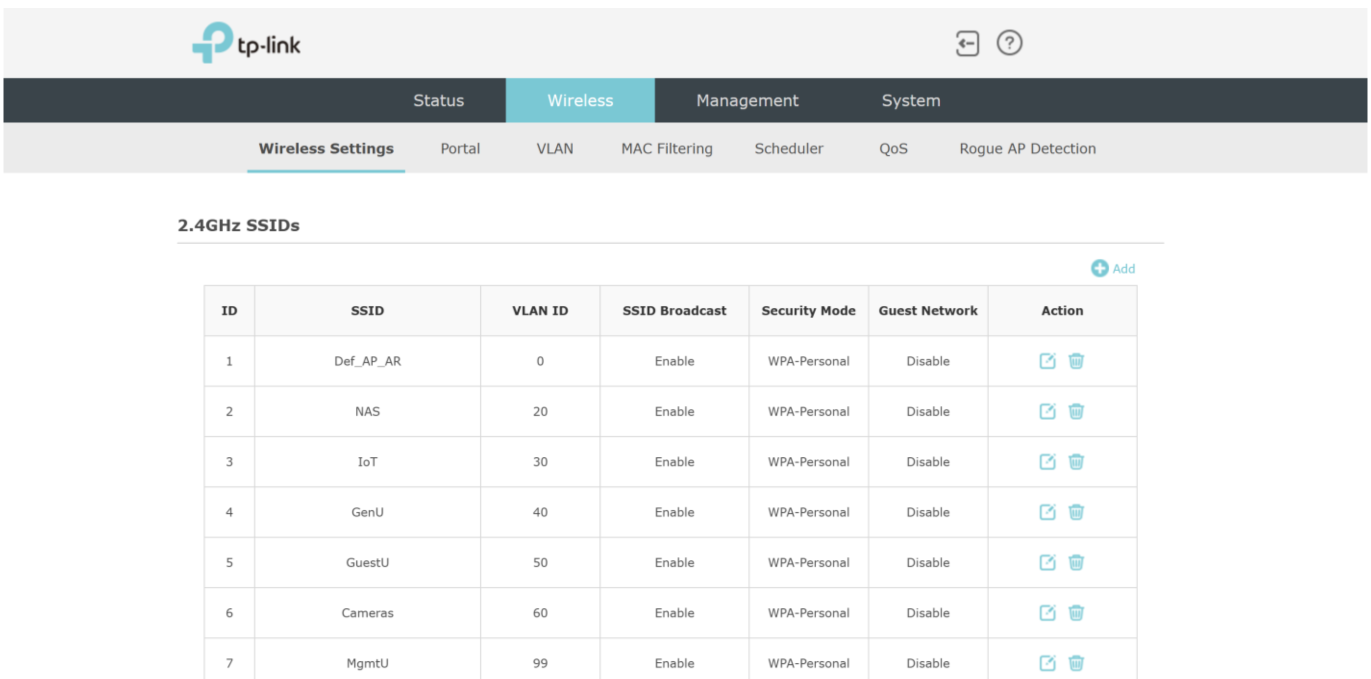


Figure 15: TP-Link AP management network page showing static IP 10.10.99.21, gateway 10.10.99.1, and DNS server 10.10.99.10.



ID	SSID	VLAN ID	SSID Broadcast	Security Mode	Guest Network	Action
1	Def_AP_AR	0	Enable	WPA-Personal	Disable	
2	NAS	20	Enable	WPA-Personal	Disable	
3	IoT	30	Enable	WPA-Personal	Disable	
4	GenU	40	Enable	WPA-Personal	Disable	
5	GuestU	50	Enable	WPA-Personal	Disable	
6	Cameras	60	Enable	WPA-Personal	Disable	
7	MgmtU	99	Enable	WPA-Personal	Disable	

Figure 16: TP-Link AP wireless VLAN page showing NAS VLAN 20, IoT VLAN 30, GenU VLAN 40, GuestU VLAN 50, Cameras VLAN 60, and MgmtU VLAN 99.

11. Firewall Policy and Security Zones

Firewall policy is applied on each pfSense interface. Rules are evaluated top-to-bottom. Specific allow and block rules must appear above broad allow rules. The goal is to allow DNS and required services first, block access to management or internal private networks where needed, and then allow internet access only where appropriate.

11.1 Firewall aliases

Alias	Type	Members	Purpose
DNS_SERVER	Host	10.10.99.10	Single reference for internal DNS rules
MANAGEMENT_HOSTS	Hosts	10.10.99.1, 10.10.99.10, 10.10.99.20, 10.10.99.21, 192.168.1.50	Infrastructure devices protected from normal/guest/IoT access
MGMT_PORTS	Ports	22, 80, 443, 8006, 5380	Common administrative ports for SSH, web interfaces, Proxmox, and Technitium
RFC1918_PRIVATE	Networks	10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16	Used to block internal/private network access while still allowing internet

11.2 Rule model by interface

Interface	Rule order	Purpose
WAN	No public management rules; no port forwards for admin services	Minimizes inbound exposure from the upstream/internet side
Tailscale	Allow Tailscale to MANAGEMENT_HOSTS on MGMT_PORTS; allow DNS to DNS_SERVER; deny other access unless needed	Remote admin access without advertising every VLAN
VLAN99_MANAGEMENT	Allow admin access to MANAGEMENT_HOSTS; allow DNS; allow internet/general admin traffic	Trusted local admin VLAN
LAN / VLAN1	Allow DNS; block MANAGEMENT_HOSTS on management ports; allow general access if still required	Keeps native LAN usable without making it the management zone
NAS_V20	Allow DNS; block MANAGEMENT_HOSTS; allow required internet/updates	Storage segment protected from admin devices
IOT_V30	Allow DNS; block RFC1918_PRIVATE; allow internet	IoT can reach internet but not internal private networks
NORMALUSERV40	Allow DNS; block MANAGEMENT_HOSTS; allow internet/general access	Normal users cannot manage infrastructure
GUESTUSERV50	Allow DNS; block RFC1918_PRIVATE; allow internet	Guests are isolated from all private internal networks
VLAN60_CAMERAS	Allow DNS; allow NTP; block RFC1918_PRIVATE; no broad internet allow unless required	Camera network is local-first and restricted

11.3 Rule-order design notes

- On every VLAN interface, DNS allow and block rules are placed above broad allow rules, because a default allow rule above a block rule would otherwise override the block.
- For each VLAN client rule, the source is set to the interface's "subnets" entry so that every client on that VLAN is covered. The "address" source option, which matches only the pfSense interface IP itself, is not used for client-access rules.
- The block rule on NAS_V20 toward MANAGEMENT_HOSTS prevents NAS-side devices from initiating connections to management devices. It does not constrain the reverse direction; management VLAN clients can still reach the NAS unless a corresponding outbound rule is added on VLAN99.
- From VLAN99 the pfSense GUI is also reachable on any of its other interface addresses (for example 10.10.20.1 or 10.10.30.1), since they all resolve to the same pfSense instance. https://10.10.99.1 is used as the canonical management URL across the documentation.

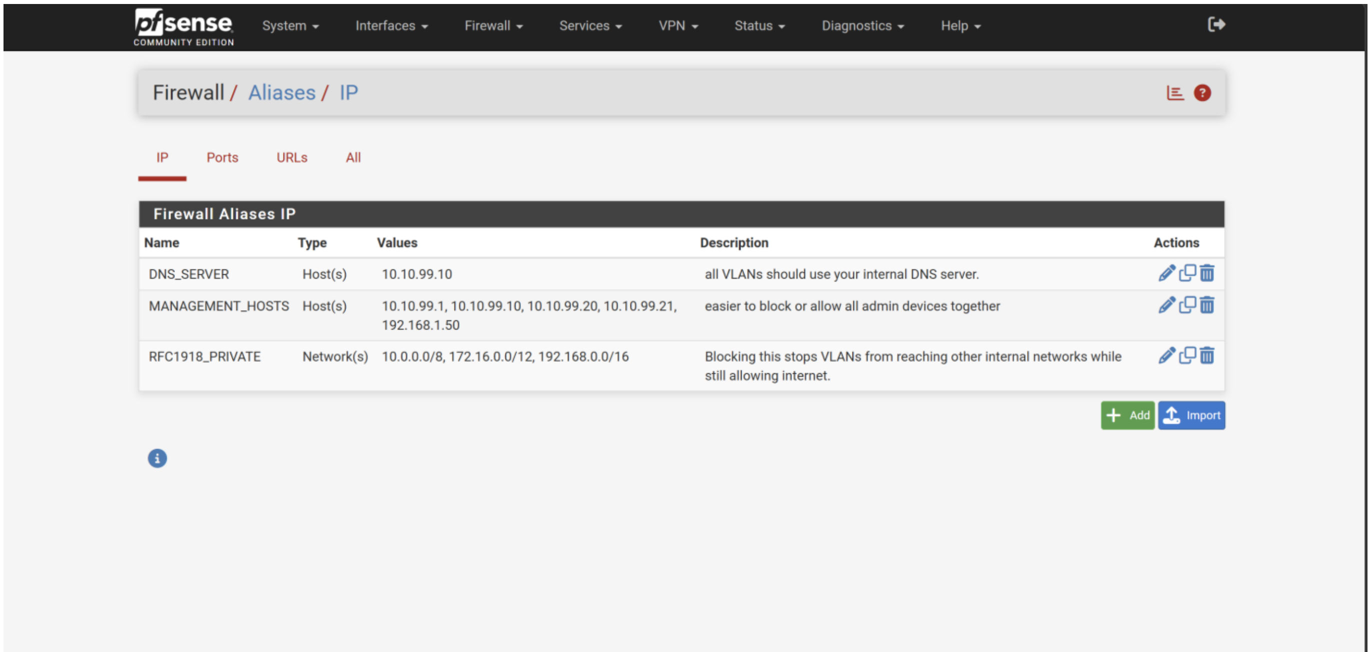


Figure 17: pfSense firewall aliases page showing DNS_SERVER, MANAGEMENT_HOSTS, MGMT_PORTS, and RFC1918_PRIVATE.

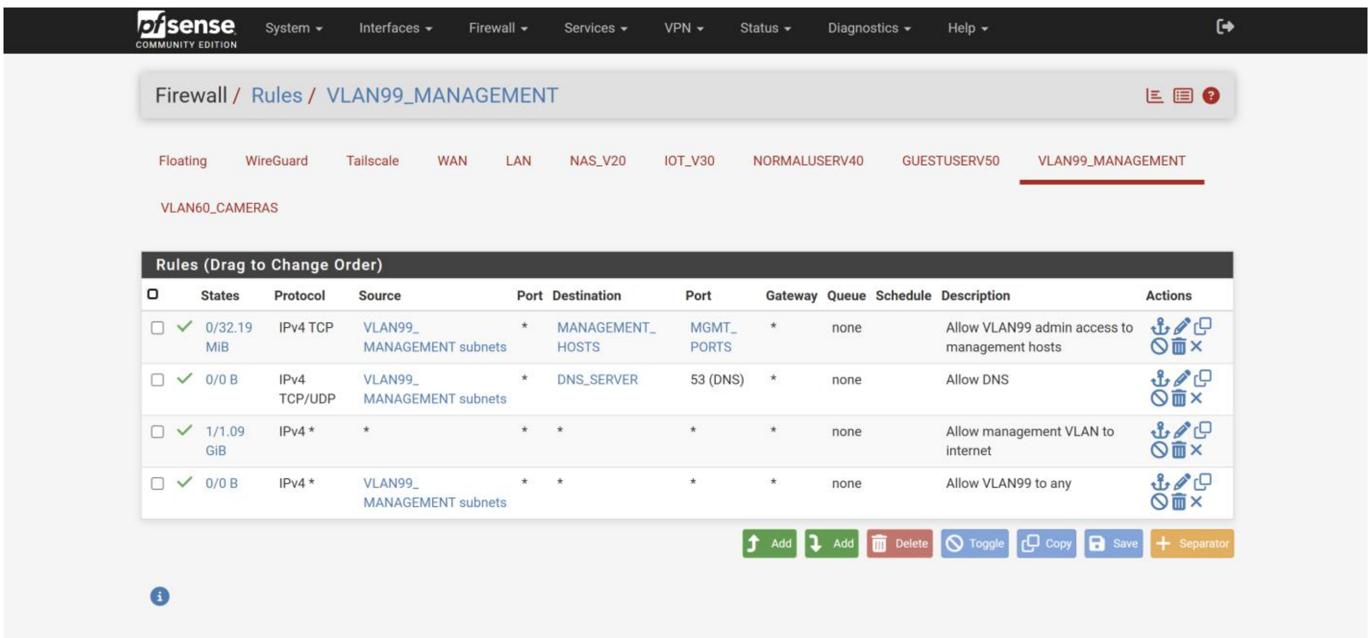


Figure 18: pfSense VLAN99_MANAGEMENT rules showing management access to MANAGEMENT_HOSTS, DNS access to DNS_SERVER, and controlled internet/general access.

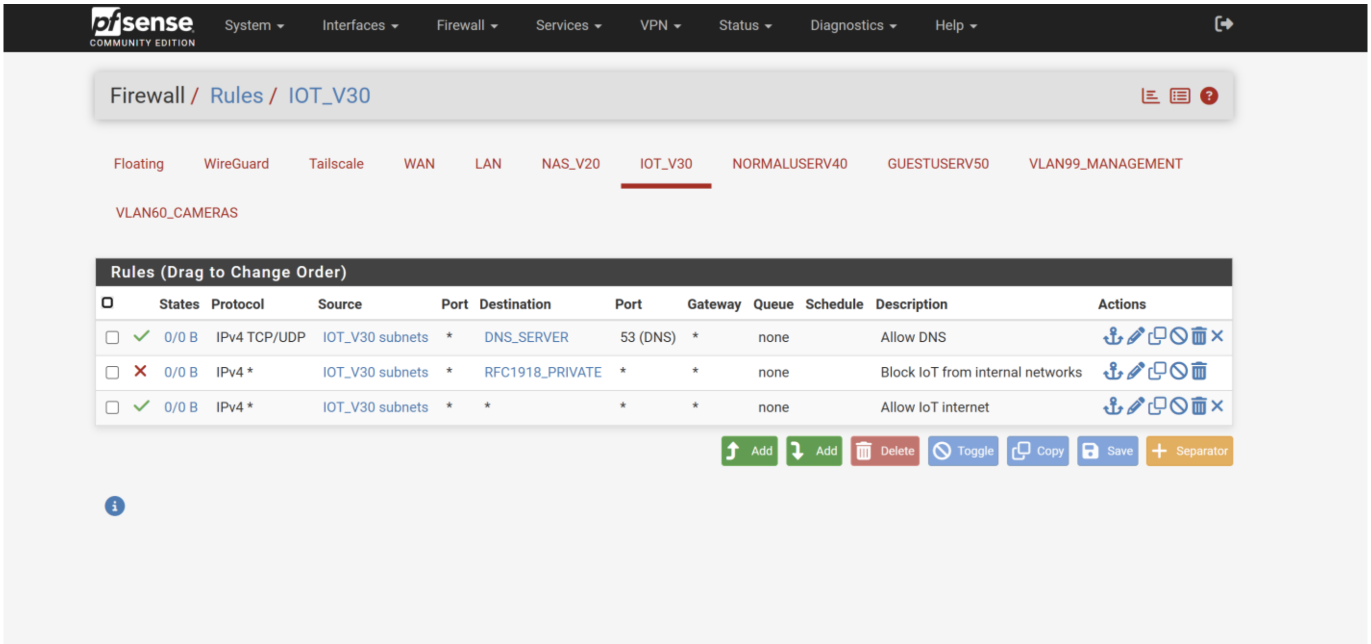


Figure 19: IOT_V30 rules showing DNS allow, RFC1918_PRIVATE block, and internet allow after the block rule.

12. Validation and Testing Procedure

Validation was performed from both local VLAN clients and from a remote Tailscale-connected laptop. The test plan covered addressing, DNS resolution, internet access, firewall isolation between zones, and remote administration through the tailnet. All tests returned the expected results, confirming the design works end-to-end without any reliance on public port forwarding.

12.1 Local client tests

Test	Command / action	Result
DHCP lease on management VLAN	ipconfig /all	Client receives 10.10.99.x, gateway 10.10.99.1, DNS 10.10.99.10
DNS server reachability	ping 10.10.99.10	Replies from DNS server
Internal DNS resolution	nslookup pfsense.arif.local 10.10.99.10	Resolves to 10.10.99.1
External DNS resolution	nslookup google.com 10.10.99.10	Returns public Google addresses
Internet connectivity	ping 8.8.8.8	Replies indicate routing/NAT works
Management access	Open https://10.10.99.1, http://10.10.99.20, http://10.10.99.21	pfSense, switch, and AP pages load from VLAN99

12.2 Tailscale remote tests

Test	Command / action	Result
Tailscale status	tailscale status	Laptop and pfSense appear connected in the same tailnet
Remote pfSense access	Open https://10.10.99.1	pfSense login page loads through Tailscale
Remote DNS GUI	Open http://10.10.99.10:5380	Technitium interface loads
Remote switch access	Open http://10.10.99.20	Switch interface loads
Remote AP access	Open http://10.10.99.21	AP interface loads; some AP UI pages may be slower over Tailscale
Remote Proxmox access	Open https://192.168.1.50:8006	Proxmox GUI loads through the advertised 192.168.1.0/24 route

A negative test was also run by stopping the Tailscale client on the remote laptop (tailscale down). With the laptop physically off VLAN99 and Tailscale stopped, the management addresses 10.10.99.1 and 10.10.99.20 were no longer reachable, confirming that remote access depends entirely on the tailnet rather than on any public exposure of the lab network.

13. Backup and Recovery Procedure

Configuration backups were taken once the firewall, VLAN, switch, AP, and Tailscale configuration reached a stable state. The environment has several interdependent components, and a misconfiguration on a VLAN trunk, a pfSense rule, or a management IP can lock out further access; the backups provide a known-good recovery point for each device.

13.1 Completed backups

Device / service	Backup file example	Backup method	Notes
pfSense	config-pfSense.home.arpa-YYYYMMDDHHMMSS.xml	Diagnostics > Backup & Restore > Download configuration	Most important backup; includes interfaces, DHCP, aliases, firewall rules, and package settings
TP-Link switch	SWITCHconfigfile.json	System Tools > Backup / Restore	Contains VLAN table, PVIDs, management IP, and port settings
TP-Link AP	APconfig.bin	System > Backup & Restore	Contains AP management settings and SSID/VLAN mapping
Technitium DNS	Zone export / configuration backup	Technitium web UI backup	Contains DNS info

13.2 Backup storage and handling

- One copy of each backup file is kept on the administrator laptop in a clearly named project folder.
- A second copy is kept on an external drive and in cloud storage as an off-host recovery copy.
- The backup files contain device credentials, authentication keys, and account identifiers, so they are treated as private artefacts and are not distributed alongside the documentation.
- The associated documentation and the sanitised diagrams are the artefacts intended for shared distribution; the live device backup files remain separate.

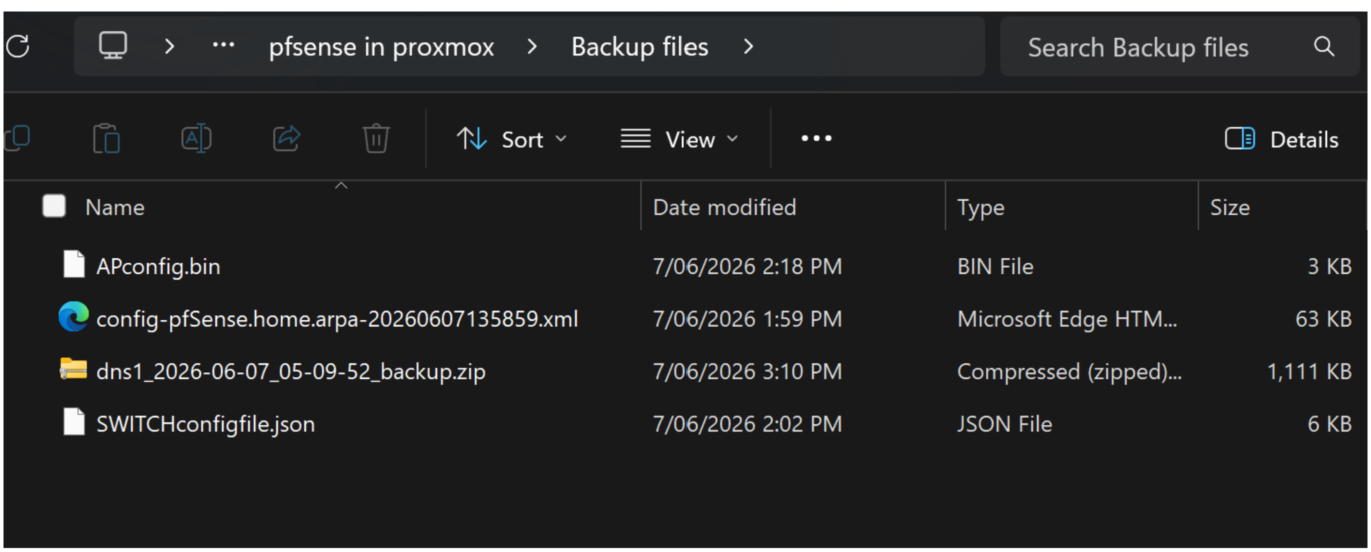


Figure 20: Backup folder showing pfSense XML backup, switch JSON backup, AP BIN backup, and optional Technitium export file.

14. Troubleshooting Notes

The table below records issues encountered during the build and the resolutions that were applied. It is included as part of the implementation record and helps explain several of the design decisions captured in earlier sections.

Issue	Likely cause	Resolution / lesson
pfSense could not be reached from 192.168.1.x	Access attempt was from pfSense WAN side, where pfSense blocks unsolicited management access by default	Use LAN/VLAN99 for management, or use Tailscale. Do not rely on WAN GUI access.
VLAN interface accidentally assigned to WAN	Management VLAN was mistakenly placed on the WAN parent/interface	Reassign VLANs to the LAN parent/trunk interface and verify via console/interface list.
Ubuntu DNS showed loopback resolver 127.0.0.53	Ubuntu systemd-resolved local stub resolver was being queried from the host itself	Test using explicit DNS server: nslookup google.com 10.10.99.10, and verify Technitium listening and forwarding.
Internet failed on internal clients	Firewall/NAT/DNS path incomplete or wrong rule order	Verify pfSense WAN gateway, NAT outbound, DNS forwarding, and VLAN rules.
AP web interface loads slowly over Tailscale	AP web UI behaviour over routed/VPN path; routing is working if other devices load normally	Use local VLAN99 for AP changes when needed; keep AP firmware and browser cache clean.

15. Outcome

This implementation delivers a complete segmented network built from a small set of cooperating components: Proxmox VE for virtualization, pfSense for firewalling, routing, NAT, and DHCP, Technitium DNS on Ubuntu Server for internal name resolution, a TP-Link managed switch for VLAN trunking, a TP-Link access point for wireless VLAN mapping, and Tailscale for remote administration. Because Tailscale establishes outbound connections from both endpoints into the tailnet, remote access works without any inbound port forwarding on the upstream router and stays independent of the public IP.

The build applies enterprise-style segmentation principles within a home-lab footprint. Infrastructure devices are grouped in a dedicated management VLAN, untrusted device classes are placed on independent VLANs with their own firewall policies, DNS is resolved through a single internal server rather than per-device upstream resolvers, and remote administration runs over an authenticated Tailscale overlay instead of any directly exposed service on the public internet. Each layer reinforces the others, and the firewall, DNS, switch, AP, and Tailscale configurations have all been backed up at a known-good state.

Throughout this document, credentials, authentication keys, public IP details, device serial numbers, and private account identifiers have been omitted from both the prose and the accompanying screenshots. The content focuses on the addressing plan, VLAN structure, firewall logic, and remote-access configuration needed to reproduce the architecture.